TERRA CLASSIC CORE

PAY PER JOB: SOLIDPROOF SECURITY AUDIT

[Text Proposal for Discussion] Pay Per Job: SolidProof Security Audit of the Terra Classic Core Chain

Prepared by: Terra Classic Core, a coalition of Bumeo Capital and LunaClassicDAO
Proposers: Puya (Bumeo Capital) and Classic DAO (LunaClassicDAO)
Named partners: Renzo (LunaClassicDAO) and Till, also known as Frag (Bumeo Capital)
Type: Text proposal under the Pay Per Job structure. This vote moves no funds.
Auditor: SolidProof (FutureVisions Deutschland UG)
Agreement: Security Audit Agreement dated 28 October 2026, fixed fee USD 85,200
Amount: No LUNC amount is set in this text vote. Each spend is sized at the LUNC rate when that spend is submitted.

SUMMARY

This text proposal asks the community to approve one job: an independent security audit of the Terra Classic core chain by SolidProof, paid under the Pay Per Job structure.

The engagement covers the live mainnet core at tag `v4.0.1-patch.3`, plus the pre-deployment work on the `ustc_staking` branch: **Market Module 2.0**, the new **x/ustcstaking** (USTC staking) module, the tax-redirect path, and the **v15** upgrade handler. The full fee is fixed at **USD 85,200** for **568 engineer-hours**, including a full re-audit after remediation. Remediation itself is done by Terra Classic Core developers at no cost to the community pool.

The job is funded in two steps, each with its own community pool spend proposal after this text passes:

1. **40% upfront (USD 34,080)**, paid before SolidProof starts work, under its own spend proposal.
2. **60% on delivery of findings (USD 51,120)**, under its own spend proposal. The re-audit is included in this part and is not charged separately.

Each spend's LUNC amount is sized at the LUNC rate at the time that spend is submitted, so SolidProof receives the contracted USD amount after fees. The LUNC is converted off-chain through SimpleSwap (about 2–2.5% on recent quotes at each 40%/60% tranche size), and moving LUNC on-chain incurs about 1.5% in Terra Classic fees. Together that is about 3.5–4% over the contracted USD on each spend. Any leftover funds are returned to Terra Classic Core.

Classic DAO (LunaClassicDAO) and Bumeo Capital commit to fixing all findings from SolidProof's audit **free of charge**, through Renzo (LunaClassicDAO) and Till, also known as Frag (Bumeo Capital), working jointly. The community pays only SolidProof's audit fee and the conversion cost.

- **Scope:** Core protocol (custom modules, SDK overrides, fees/tax, app and IBC wiring, CosmWasm), SDK53 migration and state integrity, adversarial and economic review, Market Module 2.0, x/ustcstaking, v15 upgrade, and one full re-audit after remediation
- **Audit commit:** `35d68fe1b3eb811c27e8bf3f398d5cc197897229` on branch `ustc_staking` (2026-10-15), based on `f54cdec` (`v4.0.1-patch.3`) plus 52 commits
- **Timeline:** About 8 weeks of active auditor work to delivery of findings (after a 3-week lead time from receipt of the upfront payment), then a Client remediation window of up to 4 weeks, then re-audit and Final Audit Report
- **Deliverables:** Threat model, interim findings, draft audit report, public status notes for governance, Final Audit Report after re-audit, recommendation summary, hardened CI config, fix verification statement
- **Total price:** USD 85,200 (fixed)
- **Development / fix cost:** USD 0

RATIONALE

The core chain carries Terra Classic's monetary and fee logic, including treasury, tax and tax exemption, oracle, market, dynamic commission and vesting, plus the app wiring and CosmWasm bindings. The SDK53 upgrade moved the chain across Cosmos SDK 0.47 to 0.53, ibc-go v7 to v10, and from the classic-terra wasmd fork to upstream wasmd 0.61. That code is live on mainnet and has not been independently audited.

Separately, Market Module 2.0 and USTC staking are being prepared on the `ustc_staking` branch for the v15 network upgrade. Those additions move funds and re-activate swap economics, so they should be reviewed before deployment.

An independent audit of this frozen commit gives validators, developers and holders a public, expert review of both the live core and the pre-deployment modules. The partnership keeps the cost to the community pool to the audit fee and the conversion cost. SolidProof finds and reports the issues. Renzo and Till (Frag) fix them at no charge.

This proposal follows the Pay Per Job process: the job is approved by a text vote first, and money moves only through separate spend proposals tied to the work.

SCOPE OF WORK

Per the SolidProof Security Audit Agreement dated 28 October 2026 (which supersedes earlier drafts scoped against the pre-SDK53 codebase):

Audit target

- Repository: https://github.com/classic-terra/core
- Branch: `ustc_staking`
- Frozen commit: `35d68fe1b3eb811c27e8bf3f398d5cc197897229` (2026-10-15)
- Base: `f54cdec2ed8aa04a83d904fe21d5d2a63d35ff93` (`v4.0.1-patch.3`, live on mainnet) plus 52 commits
- Size at freeze: 30,329 raw lines / 23,203 nSLOC of non-test, non-generated Go (plus 31 `.proto` definitions in scope)

In scope (contracted workstreams)

Workstream	Hours	Fee (USD)
Core protocol audit (custom modules, fees/tax, SDK overrides, CosmWasm, IBC, node surface)	288	43,200
Framework migration and state integrity review (SDK53, CosmWasm store migration, upgrade handlers)	88	13,200
Adversarial simulation and economic review	64	9,600
Market Module 2.0, x/ustcstaking (USTC staking), tax redirect, and v15 upgrade	96	14,400
Full re-audit after remediation	32	4,800
Total	**568**	**85,200**

Review depth is allocated by risk (D1 deep / D2 targeted / D3 tooling sweep). Market Module 2.0 and x/ustcstaking are reviewed at D1 as fund-moving code.

Explicitly included pre-deployment additions

- Market Module 2.0 re-implementation and re-activated swap economics (allowed swap denoms restricted to `uusd`)
- New `x/ustcstaking` module (staking and reward logic)
- Tax-redirect mechanism and market accumulator in the fee and burn-tax path
- Oracle TWAP and freshness hooks feeding the market
- v15 upgrade handler

Out of scope (per the agreement)

- Upstream Cosmos SDK, CometBFT, ibc-go, wasmd and wasmvm source, except where Terra Classic modifies or configures them
- Deployed CosmWasm contracts (chain-side integration only)
- Generated protobuf bindings
- Off-chain infrastructure (validators, relayers, RPC, explorers, wallets, price feeders)
- Formal verification
- Economic policy choices (mechanism implementation is in scope; parameter advisability is not)
- Development landing after the freeze, other than the remediation diff verified in the re-audit

Changes after the freeze are a change order. The agreement also lists optional add-ons (depth upgrades, further re-audit pass, exhaustive store-migration replay) that would need their own spend if chosen later.

Deliverables

- Threat Model and Architecture Note (early)
- Hardened continuous integration configuration (pull request early in the engagement)
- Interim Findings Report (Critical and High findings mid-engagement)
- Audit Findings and Draft Audit Report (triggers the 60% payment)
- Public Audit Status Note (twice: at start with the 40% spend, and on delivery of findings with the 60% spend)
- Final Audit Report after re-audit (publishable)
- Recommendation Summary (consensus-breaking vs non-breaking)
- Fix Verification Statement (per finding, as part of the re-audit)

SolidProof reports Critical findings within 24 hours and High findings within three business days. Unfixed Critical and High findings stay under embargo until a patch is adopted or 90 days, whichever is earlier.

Entry Criteria (Before Work Starts)

- This text proposal passes.
- The SolidProof Security Audit Agreement dated 28 October 2026 (identified by its SHA-256 hash in the on-chain text) is the accepted statement of scope, commit, timeline, deliverables and price.
- The first spend proposal (40% upfront, USD 34,080) passes and the funds are converted and paid to SolidProof in USDC.
- Renzo (LunaClassicDAO) and Till (Frag, Bumeo Capital) confirm they are available to fix findings at no cost to the community pool.

Exit Criteria (Completion)

- SolidProof delivers the Audit Findings and Draft Audit Report.
- The second spend proposal (60% on delivery of findings, USD 51,120) passes and the funds are converted and paid to SolidProof in USDC.
- Renzo and Till (Frag) fix the findings jointly, at no cost.
- SolidProof completes the included re-audit and issues the Final Audit Report and Fix Verification Statement.

- The audit report and fix status are posted in this Agora thread.

AMOUNT AND PAYMENT

Rate: USD 150 per engineer-hour. Fixed total: USD 85,200 for 568 hours. No project fee, management surcharge or tooling licence charge beyond that rate.

Item	Share	Amount (USD)
Upfront payment, before SolidProof starts (own spend proposal)	40%	34,080
Payment on delivery of findings, including the re-audit (own spend proposal)	60%	51,120
Fixing all audit findings (Renzo and Till (Frag), for Classic DAO (LunaClassicDAO) and Bumeo Capital)		0
Development		0
Total	**100%**	**85,200**

Payment structure (as set in §14 of the SolidProof Security Audit Agreement dated 28 October 2026):

- **Client.** Terra Classic (the chain) is not a legal person. The Client under the agreement is **Terra Classic Core** (Bumeo Capital and LunaClassicDAO), acting through its authorised representatives Puya (Bumeo Capital) and Classic DAO (LunaClassicDAO). Both payments under the 40/60 split are made only if their community-pool spend proposals pass.
- **No LUNC amount in this vote.** This text proposal approves the job and the payment structure only. It does not move any funds or fix any LUNC amount.
- **Two spends, each sized at its own time.** Each spend proposal requests the LUNC needed for its share at the LUNC rate at the time that spend is submitted. The rate source, date and resulting LUNC and uluna amount will be posted with each spend.
- **No buffer.** No 20% buffer or any other buffer is added on top of the price.
- **No prepaid fix budget.** Fixes are done by Renzo and Till (Frag) at no cost, so no money is set aside for fixes.
- **Fee fixed in USD, paid in USDC.** SolidProof bears no LUNC price, conversion or slippage risk. The fee is remitted in USDC to SolidProof's address in Appendix D of the agreement.
- **Funds flow.** Each spend pays from the community pool to the Client receiving multisig. The multisig converts LUNC off-chain through SimpleSwap (about 2–2.5% on recent quotes at each 40%/60% tranche size) and pays SolidProof in USDC. Moving LUNC on-chain also incurs about 1.5% in Terra Classic fees. Together that is about 3.5–4% over the contracted USD. Those costs are borne by the community pool; each spend is sized so SolidProof still receives the contracted USD amount after fees.
- **Leftovers.** Any LUNC or converted funds left after SolidProof is paid are returned to Terra Classic Core, and the transaction is posted in this thread.
- **Receiving multisig.** Four signers: Renzo (LunaClassicDAO), Frag (Bumeo Capital), Puya (Bumeo Capital) and Classic DAO (LunaClassicDAO). Signing threshold: _____. **The multisig address will be posted in this thread when this text proposal is up for voting.**
- **SolidProof payment address and currency:** USDC, address per Appendix D of the agreement (to be posted in this thread before the first spend).
- **On-chain citation.** The on-chain text of this proposal will cite the agreement by its SHA-256 hash so the scope, commit, timeline, deliverables and price stay fixed.

VAT: statutory VAT applies only if no VAT-ID is provided; it is not applicable if the customer is a registered business outside the European Union.

TIMELINE

SolidProof starts only after this text proposal passes and the 40% upfront payment has been received. There is a three-week lead time from receipt of that payment. Timings below are measured from commencement of auditor work.

When	Milestone
After this text passes	First spend proposal (40%, USD 34,080) submitted and voted
After the first spend passes and payment is received	Three-week lead time, then SolidProof starts
End of week 1 from start	Threat Model and Architecture Note; hardened CI pull request
End of week 5 from start	Interim Findings Report
End of week 8 from start	Audit Findings and Draft Audit Report (triggers the 60% spend)
After delivery of findings	Second spend proposal (60%, USD 51,120) submitted and voted; funds converted and paid
Client remediation window	Up to four weeks; Renzo and Till (Frag) fix findings at no cost
Within one week of written "remediation complete"	Re-audit and Fix Verification Statement
Within five business days of the re-audit	Final Audit Report
Completion	Report and fix status posted on Agora; any leftovers returned to Terra Classic Core

Active auditor effort is about 8 weeks to delivery of findings, plus the included re-audit. Elapsed time to the Final Audit Report is about eleven to twelve weeks of active work if remediation takes about two weeks, plus the time taken by the two spend-proposal votes. Extra remediation weeks extend the calendar but not the contracted fee.

WHAT THIS PROPOSAL DOES NOT DO

- It does not move any funds. Funds move only through the two spend proposals described above.
- It does not set a LUNC amount.
- It does not fund any development or prepaid fix budget.
- It does not change any chain parameters.
- It does not approve any chain software upgrade. Any upgrade that includes audit fixes or that ships Market Module 2.0 / USTC staking / v15 would follow its own process.
- It does not deploy Market Module 2.0, x/ustcstaking or v15. Those remain pre-deployment until a separate upgrade process.

HOW THIS DIFFERS FROM PROPOSAL 12228

Proposal 12228 ("Fund Terra Classic Security Audit by SolidProof & Fixes by OrbitLabs") also proposed SolidProof as the auditor. This proposal uses the same auditor. What changes is the structure, in response to the concerns voters raised during the 12228 vote, and the scope, which now covers the post-SDK53 core plus Market Module 2.0 and USTC staking:

	Proposal 12228	This proposal
Auditor	SolidProof	SolidProof
First step	Spend proposal, with no text vote first	Text vote first, under Pay Per Job
Payment timing	Paid upfront in full (1,597,391,830 LUNC requested)	40% upfront under its own spend, 60% on delivery of findings under its own spend
Buffer	20% buffer on the audit fee	No buffer
Fixing findings	OrbitLabs, US$20,000 budget	Renzo and Till (Frag), US$0, no prepaid fix budget
LUNC amount	Set in the proposal	Not set in this text vote; each spend sized at the rate at that time
Leftover funds	Unused buffer returned	Any leftovers returned to Terra Classic Core
Scope	Pre-SDK53 / earlier SolidProof scoping	Post-SDK53 core + Market Module 2.0 + USTC staking + v15, frozen at `35d68fe`
Total cost to the community pool	US$90,800 (US$70,800 audit fee plus US$20,000 for fixes; US$84,960 requested for the audit including the buffer)	US$85,200 audit fee only (plus about 2–2.5% SimpleSwap and about 1.5% on-chain LUNC fees on each spend, about 3.5–4% all-in)

TEAM

This document is prepared by **Terra Classic Core**, a coalition formed by Bumeo Capital and LunaClassicDAO. Both organizations have Layer 1 developers and have combined forces to build one team working on the chain.

- **Puya** (Bumeo Capital and Juris Protocol)
- **Classic DAO** (LunaClassicDAO and WESO)
- **Frag**, also known as Till (Bumeo Capital and LuncGoblins), a Layer 1 developer on the chain since 2022
- **Renzo** (LunaClassicDAO), a Layer 1 developer on the chain since 2022

Authorised representatives for the Client under the SolidProof agreement: Puya (Bumeo Capital) and Classic DAO (LunaClassicDAO). Audit findings are fixed at no cost by Renzo and Frag, working jointly.

VOTING OPTIONS

- **YES:** approve the SolidProof security audit of the Terra Classic core chain (including Market Module 2.0, USTC staking and v15 on the frozen `ustc_staking` commit) as a Pay Per Job job, at the fixed fee of USD 85,200, paid 40% upfront and 60% on delivery of findings through two separate spend proposals, each sized at the LUNC rate at its time, with findings fixed by Renzo and Till (Frag) at no cost. No funds move with this vote.
- **NO:** do not approve this job.
- **NO WITH VETO:** you consider this proposal spam or an abuse of governance. If the veto threshold is reached, the deposit is burned.
- **ABSTAIN:** you do not wish to vote for or against, but want your vote counted toward quorum.

TO BE FINALIZED

When this text proposal is up for voting

1. The Client receiving multisig address (four signers already named: Renzo, Frag, Puya and Classic DAO; signing threshold: _____).

Before the first spend proposal goes on-chain

1. The SHA-256 hash of the accepted SolidProof Security Audit Agreement (28 October 2026), so the on-chain text points at fixed terms.
2. SolidProof's USDC payment address from Appendix D of the agreement.
3. The LUNC rate reference for each spend, posted with that spend.

QUESTIONS FOR THE COMMUNITY

1. Is the 40% upfront and 60% on delivery split the right balance between paying the auditor and protecting the community pool?
2. Is SimpleSwap acceptable for the off-chain conversion, at about 2–2.5% on recent quotes for each 40%/60% spend, plus about 1.5% in on-chain LUNC fees (about 3.5–4% all-in)?
3. Any other concerns or questions for Classic DAO (LunaClassicDAO), Bumeo Capital or SolidProof.

REFERENCES

- SolidProof: https://solidproof.io/
- SolidProof Security Audit Agreement dated 28 October 2026 (attached / posted in this thread)
- Proposal 12228 discussion: https://agora.terra-classic.io/t/proposal-terra-classic-security-audit-by-solidproof-fixes-by-orbitlabs/536
- Pay Per Job: Proposal 11889, https://station.terraclassic.community/proposal/columbus-5/11889
- Code: https://github.com/classic-terra/core
- Frozen audit commit: https://github.com/classic-terra/core/commit/35d68fe1b3eb811c27e8bf3f398d5cc197897229

This is a discussion draft aligned with the SolidProof Security Audit Agreement dated 28 October 2026. The Client receiving multisig address will be posted when this text proposal is up for voting. Multisig signing threshold, agreement SHA-256 and SolidProof's USDC payment address will be posted before the first spend proposal goes on-chain.